

Blazor, MAUI, WPF, or WinForms?

Picking the right .NET UI framework in the AI coding era

Mark Allibone

iDIP Solution



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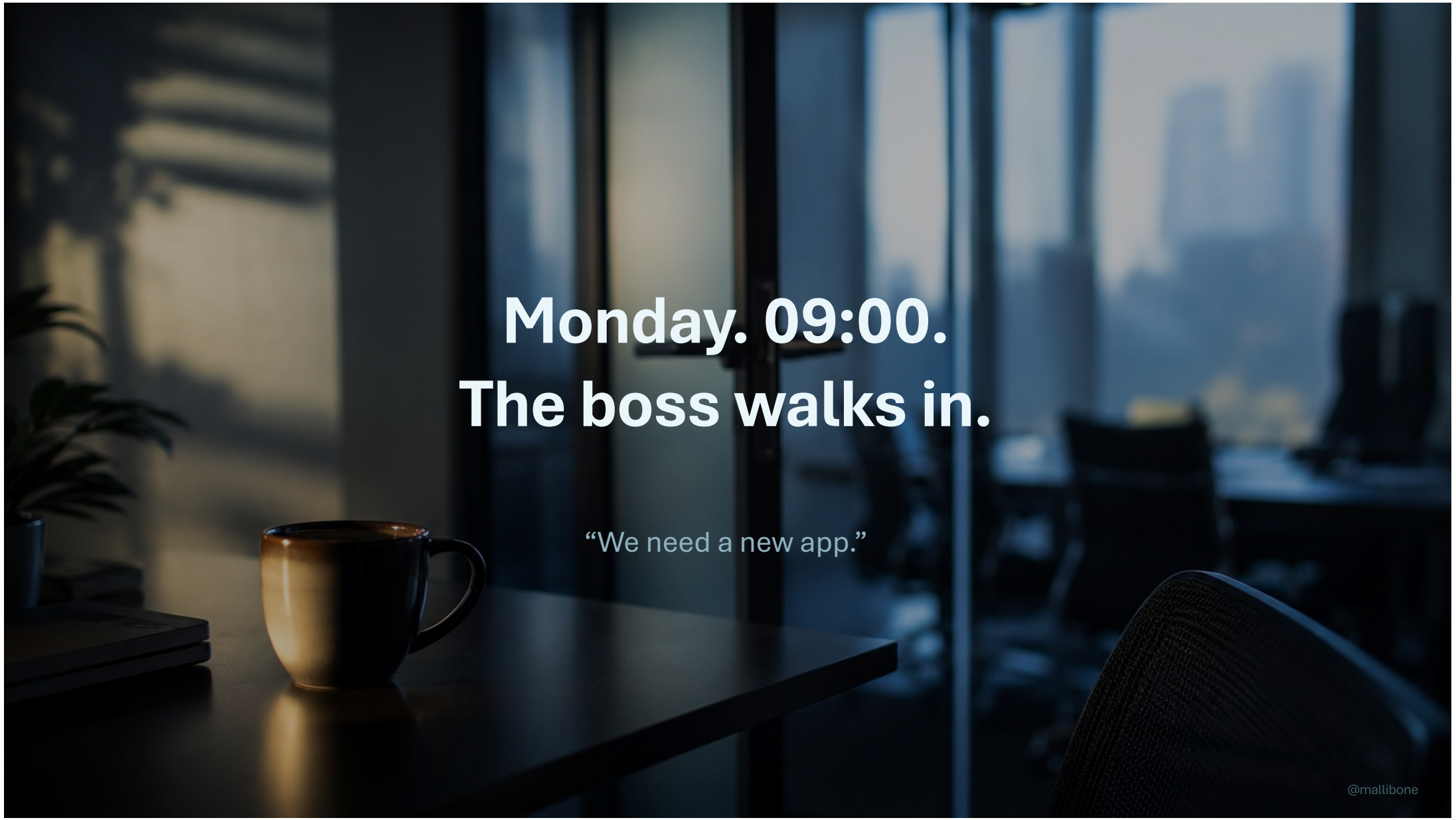
Tech Lead · Software Architect · .NET

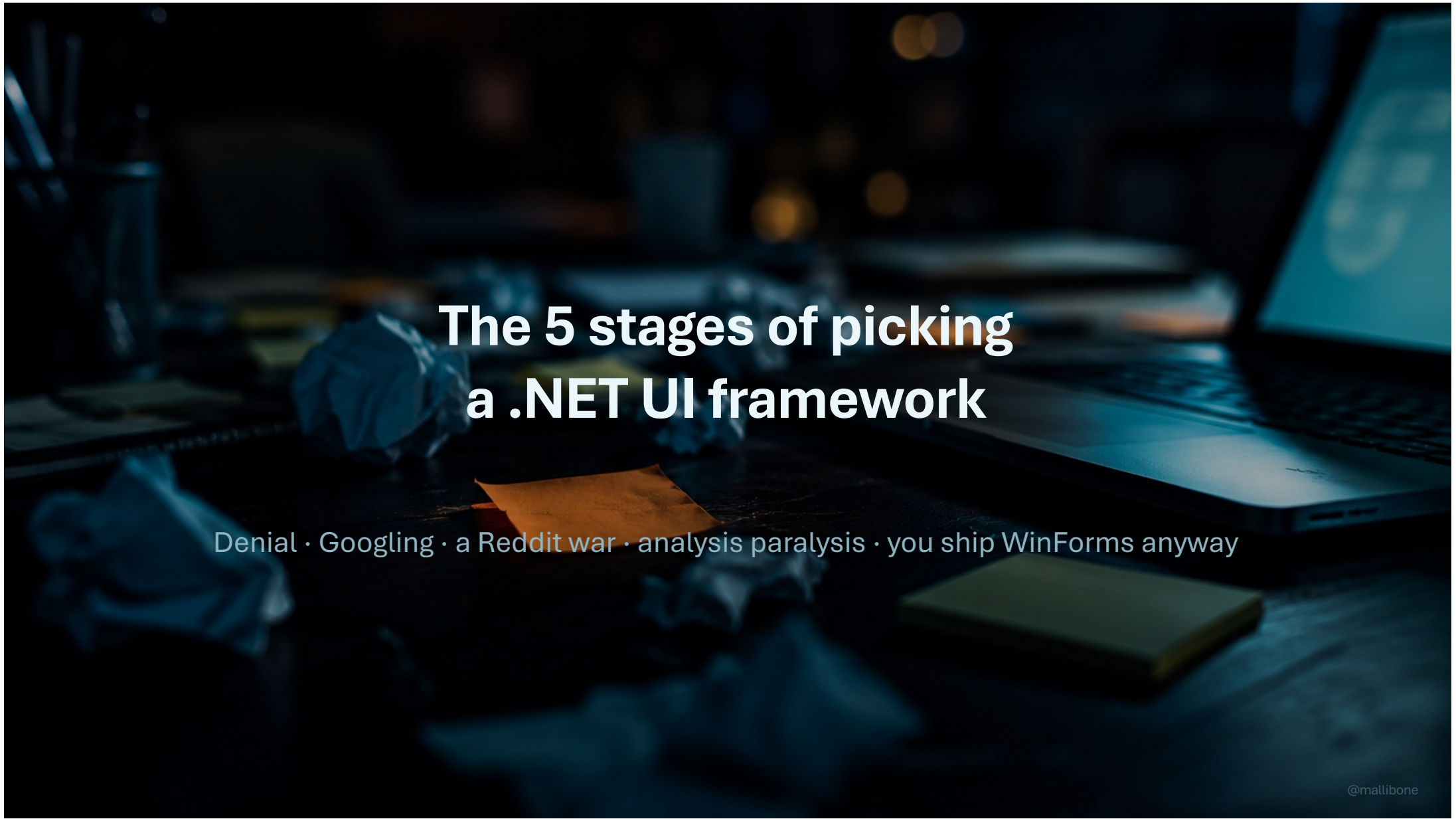
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iDIP Solution

Swiss low-code platform for industrial production

MVP Microsoft® Most Valuable Professional


Monday. 09:00.
The boss walks in.
“We need a new app.”
@mallibone


The 5 stages of picking
a .NET UI framework
Denial · Googling · a Reddit war · analysis paralysis · you ship WinForms anyway
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WHY IS THIS SO HARD?

THE CORE PROBLEM

Too many right answers

Five live, first-party .NET UI stacks in 2026. None deprecated. The 'right' one depends on requirements you don't have yet.

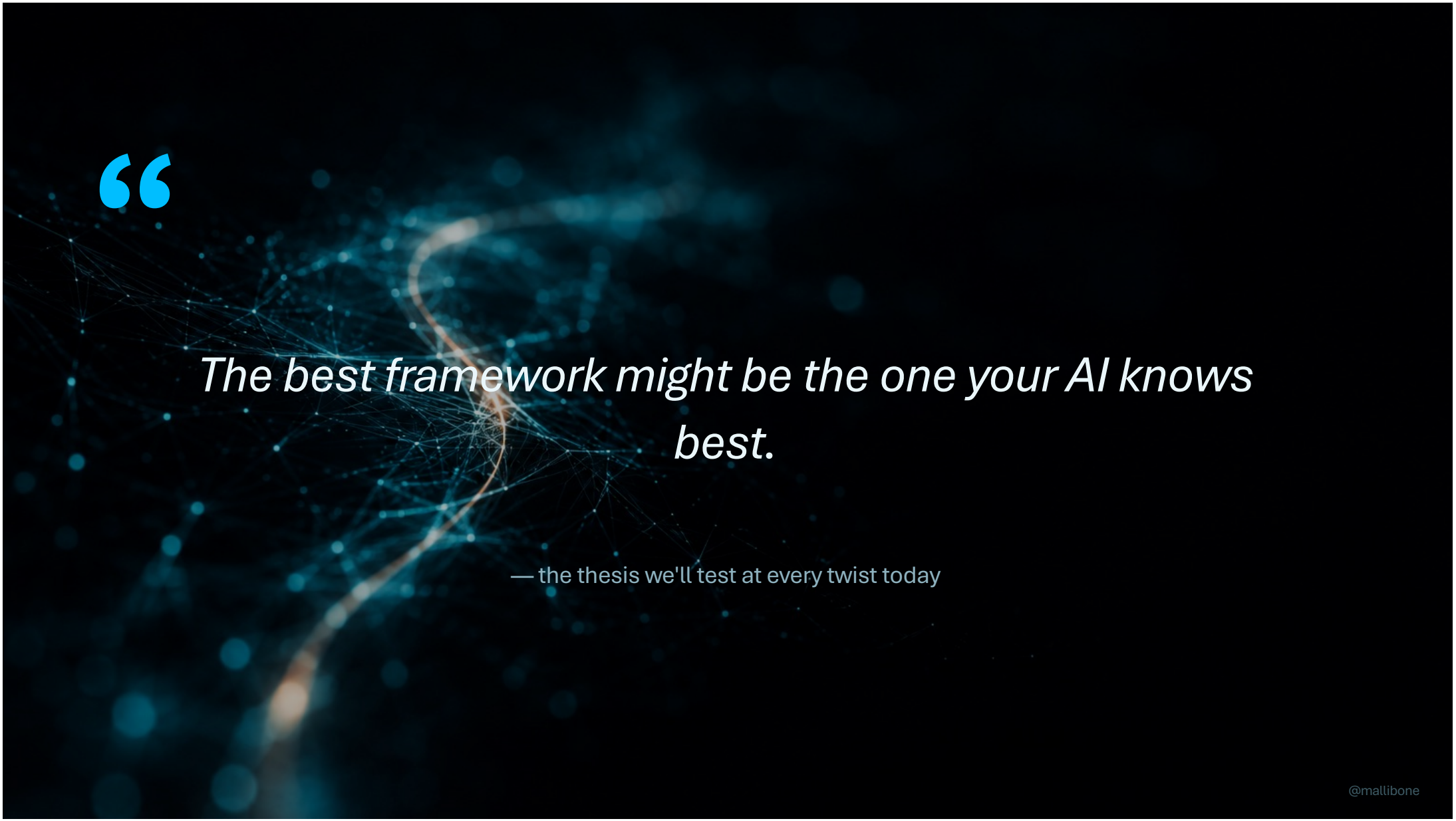

The .NET UI graveyard
Silverlight · UWP · Xamarin.Forms
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ME
SHINY
NEW FRAMEWORK
MY STABLE
APP IN PROD
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PLOT TWIST

A new axis just dropped

Copilot agent mode now writes most of the UI. The right pick now also depends on what the model writes well.


“
The best framework might be the one your AI knows best.
— the thesis we'll test at every twist today
@mallibone

HERE'S THE CATCH

The requirements never stop changing

Every few days the boss is back with one more “oh — and it also needs to…”. We start on the Windows desktop and watch the right framework change under our feet.

Six lenses

We score every framework on the same six. The last one is new this year.

1
Target platforms

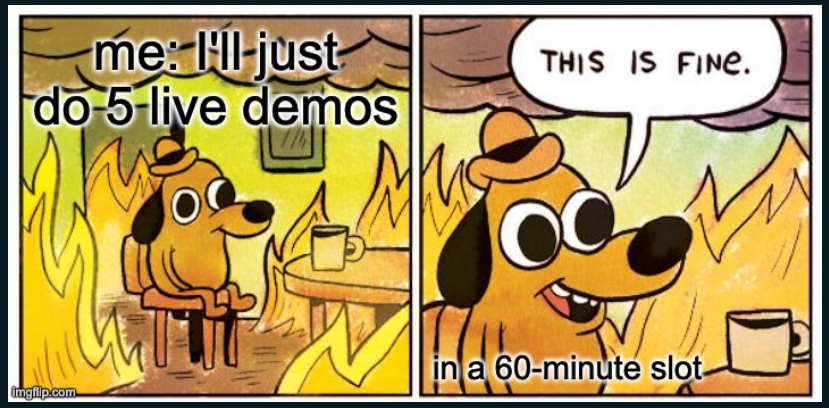
2
Developer experience

3
Ecosystem & community

4
Performance & startup

5
Distribution

6
AI / Copilot


me: I'll just do 5 live demos
THIS IS FINE.
in a 60-minute slot
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DX ALSO MEANS YOUR TOOLCHAIN

Where do you build it?

VS (Windows only) · VS Code (Mac · Win · Linux) · Rider (cross-platform).
WinForms · WPF · WinUI need Windows; Blazor & MAUI are happy on a Mac.
Tonight's Windows demos run in a VM.

REQUIREMENT 1 / 5 · THE BOSS

“We need a little Windows tool for the shop floor — clickable by Friday.”

→ **Windows Forms** fastest RAD path to a working desktop form

DEMO 1

Windows Forms — a tool by Friday

Windows Forms — flew vs flailed

WHAT FLEW

- ✓ Great event/handler code-behind
- ✓ Fast CRUD + data-binding logic
- ✓ Knows the control model cold

WHAT FLAILED

- ✗ Can't 'see' the .Designer.cs surface
- ✗ Layout drift / overlapping controls
- ✗ Designer + AI don't speak the same language

scorecard so far

Platforms	DX	Ecosystem	Perf	Distrib.	AI
weak	strong	ok	strong	ok	ok


WINFORMS IS STILL A VALID CHOICE IN 2026. CHANGE MY MIND.
CHANGE MY MIND
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REQUIREMENT 2 / 5 · THE BOSS

“Good. Now make it look the part — and fit our existing desktop suite.”

→ **WPF** mature, MVVM, integrates, deepest ecosystem

DEMO 2

WPF — make it real (MVVM)

WPF — flew vs flailed

WHAT FLEW

- ✓ Near-complete MVVM scaffolds
- ✓ Confident bindings, commands, converters
- ✓ 20 years of Stack Overflow pays off

WHAT FLAILED

- ✗ Occasionally dated patterns (code-behind)
- ✗ Reaches for heavy legacy approaches
- ✗ Rarely uses the newest tooling

scorecard so far

Platforms	DX	Ecosystem	Perf	Distrib.	AI
weak	strong	strong	ok	ok	strong

REQUIREMENT 3 / 5 · THE BOSS

“Make it feel modern — Fluent, touch, the latest Windows look.”

- → **WinUI 3** modern native Windows, Fluent, full WinRT depth

DEMO 3

WinUI 3 — modern + Fluent

WinUI 3 — flew vs flailed

WHAT FLEW

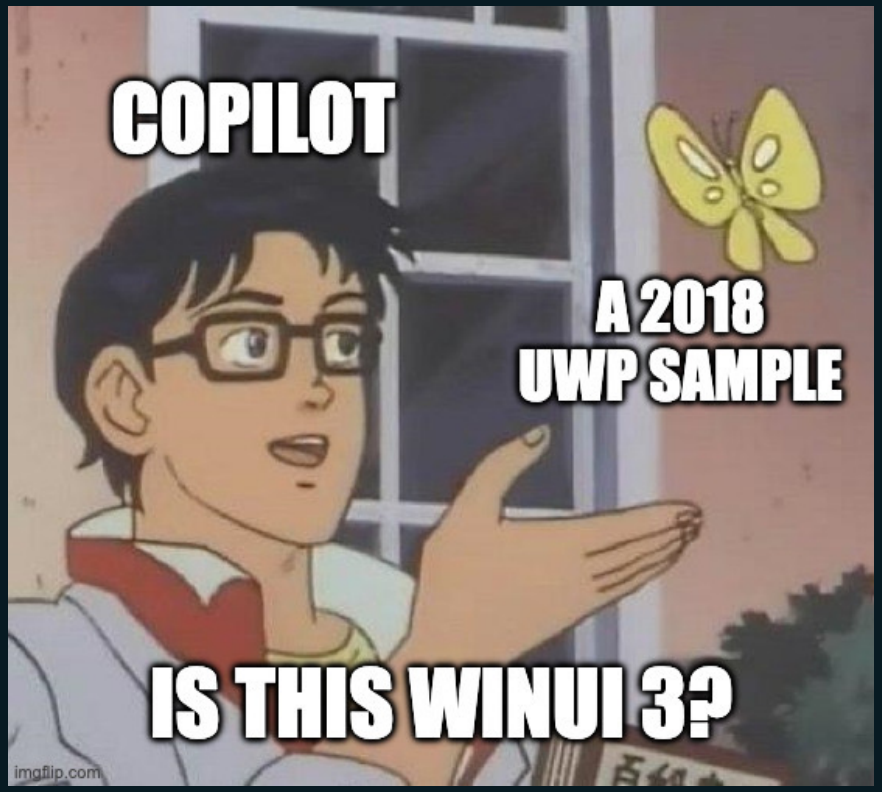
- ✓ Fluent UI looks great instantly
- ✓ Solid XAML layout suggestions
- ✓ Good x:Bind / MVVM scaffolds

WHAT FLAILED

- ✗ Conflates UWP and WinUI 3 namespaces
- ✗ Wrong using / package references
- ✗ Namespace whack-a-mole to compile

scorecard so far

Platforms	DX	Ecosystem	Perf	Distrib.	AI
weak	ok	weak	strong	ok	weak


COPILOT
A 2018
UWP SAMPLE
IS THIS WINUI 3?
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Lens 1 — Target platforms

	Windows	macOS	iOS	Android	Web
Blazor	○*	○*	○*	○*	●
.NET MAUI	●	●	●	●	—
WinUI 3	●	—	—	—	—
WPF	●	—	—	—	—
WinForms	●	—	—	—	—

● first-class ○* via Blazor Hybrid (native shell) — not a target

REQUIREMENT 4 / 5 · THE BOSS

“Oh — and the field crew needs it on their phones too.”

→ **.NET MAUI** one codebase to mobile (and desktop)

DEMO 4

.NET MAUI — onto a phone

.NET MAUI — flew vs flailed

WHAT FLEW

- ✓ Truly native controls — not a WebView
- ✓ One codebase → iOS · Android · macOS · Windows
- ✓ DevFlow: the agent sees + drives the live app

WHAT FLAILED

- ✗ Raw codegen reaches for old Xamarin APIs
- ✗ Shaky on handlers / platform specifics
- ✗ Misses newer .NET 10 properties

scorecard so far

Platforms	DX	Ecosystem	Perf	Distrib.	AI
strong	strong	ok	ok	ok	strong

MAUI · MORE THAN ONE WAY

And MAUI gives you options

XAML + MVVM · C# Markup · MVU in pure C# with MauiReactor · Blazor Hybrid.
Same app — but the AI sees very different code in each model.

MAUI · AI TOOLING

DevFlow — give the AI eyes on the app

Experimental Microsoft.Maui.DevFlow: in-app agent + CLI + an MCP server. Copilot/Claude can read the visual tree, tap, screenshot and profile a running app — and `maui devflow init` drops skills into .claude/skills/.

REQUIREMENT 5 / 5 · THE BOSS

“And customers want it in the browser — nothing to install.”

→ **Blazor** web-first, max reach, instant updates

DEMO 5

Blazor — into the browser

Blazor — flew vs flailed

WHAT FLEW

- ✓ Clean, idiomatic Razor components
- ✓ Strong web corpus → confident output
- ✓ Wired up the service unprompted

WHAT FLAILED

- ✗ Render-mode confusion: Server vs WASM vs Auto
- ✗ Prerender + interactive double calls
- ✗ Sometimes invents component params

scorecard so far

Platforms	DX	Ecosystem	Perf	Distrib.	AI
strong	strong	strong	ok	strong	strong

THE SCORECARD

The scorecard

Greener = stronger. No column wins everything — read it per lens.

	Platforms	DX	Ecosystem	Perf	Distrib.	AI / Copilot
Blazor	**Strong**	**Strong**	**Strong**	OK	**Strong**	**Strong**
.NET MAUI	**Strong**	**Strong**	OK	OK	OK	**Strong**
WinUI 3	Weak	OK	Weak	**Strong**	OK	Weak
WPF	Weak	**Strong**	**Strong**	OK	OK	**Strong**
WinForms	Weak	**Strong**	OK	**Strong**	OK	OK

AI FACTOR #1 — THE CORPUS

Training data is a feature now

Codegen quality tracks the public corpus: WPF, WinForms, Blazor rich; MAUI, WinUI thinner. But corpus isn't the whole AI story…

AI FACTOR #2 — THE TOOLING

Can your AI see the running app?

MAUI → yes: DevFlow's agent + MCP (read the tree, tap, screenshot). Web/Blazor → yes, via the browser. WPF · WinUI · WinForms → no — DevFlow is MAUI-only; just UI Automation.


“
Copilot is a brilliant senior WPF dev — from 2012.
funny, and exactly the trap
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WHICH ONE FOR YOU?

PLOT REVEAL

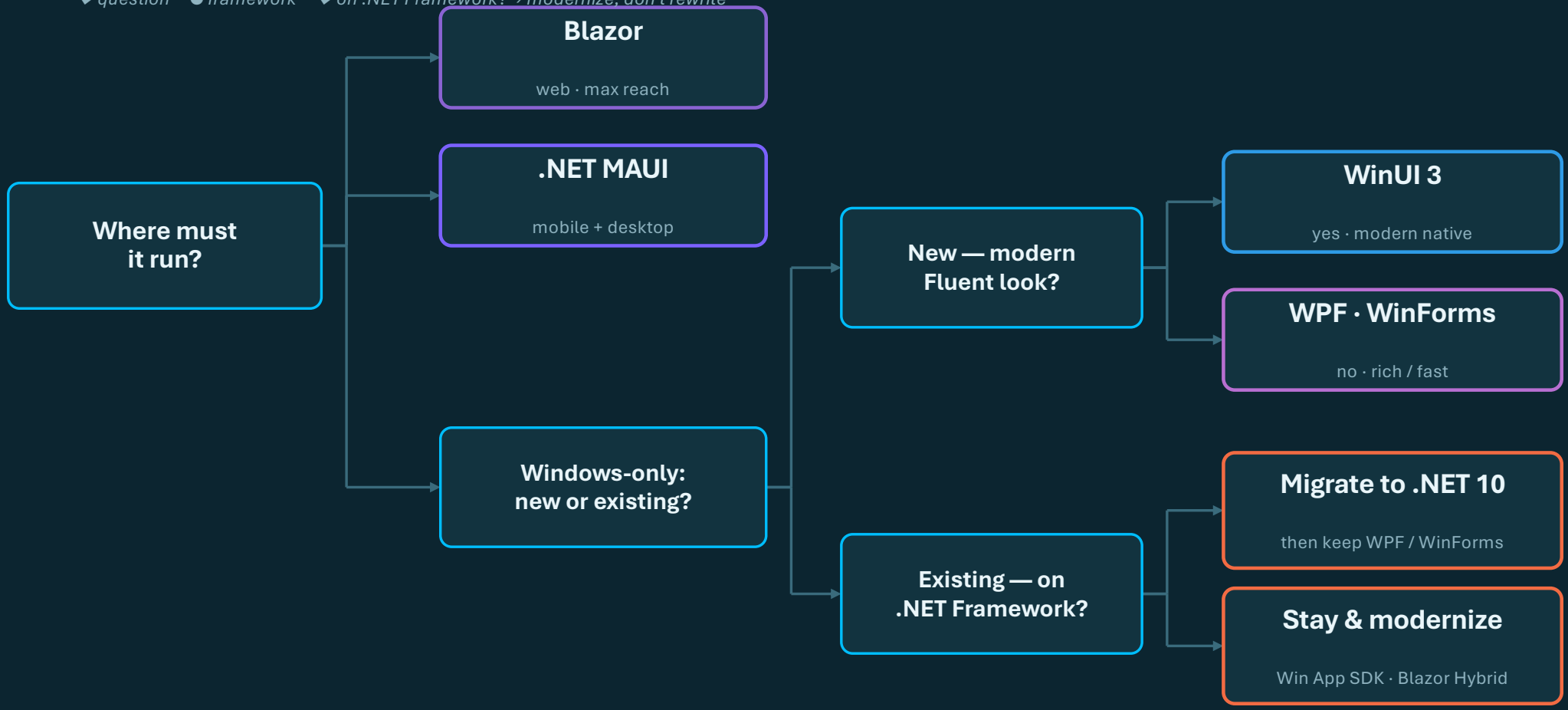
You just walked the decision tree

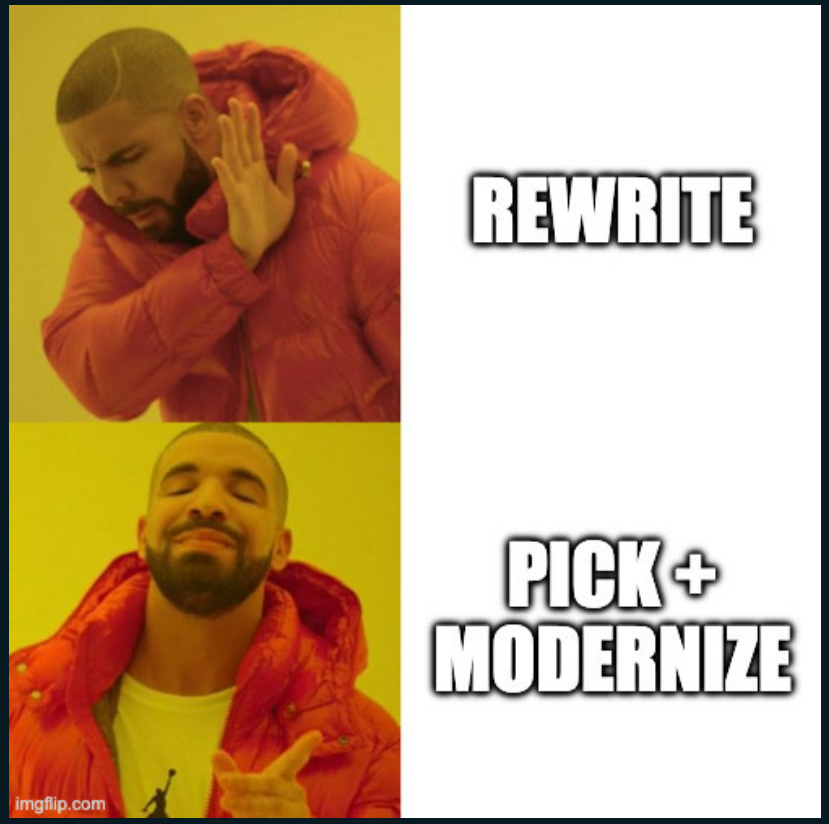
Every “oh, and it also needs to…” was a fork. We went Windows desktop → modern → multi-platform → web. The framework changed because the requirements did.


Single platform
Multi platform
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JAKE-CLARK.TUMBLR

Which one? Traverse the tree

◆ *question* ● *framework* ◆ *on .NET Framework? → modernize, don't rewrite*




REWRITE
PICK +
MODERNIZE
imgflip.com

Don't rewrite — modernize

Windows App SDK

Add Fluent, notifications, AI APIs & XAML Islands to existing WPF / WinForms — no rewrite.

Blazor Hybrid

Drop a BlazorWebView into WPF or WinForms and reuse your web components on the desktop.

Copilot app-mod agent

Automate the .NET upgrade and project-file churn with the modernization agent in VS 2026.

AND NOT ONLY THE FIRST-PARTY FIVE

On WPF and need to travel? Avalonia & Uno

Two more names worth knowing: Avalonia (incl. XPF — take an existing WPF app cross-platform) and Uno Platform (WinUI XAML everywhere, incl. the browser).

THE MINDSET

No wrong framework — only wrong reasons

Match the requirement in front of you: reach, team skills, lifespan, AI leverage. Not hype, not tribe.

THREE THINGS TO TAKE HOME

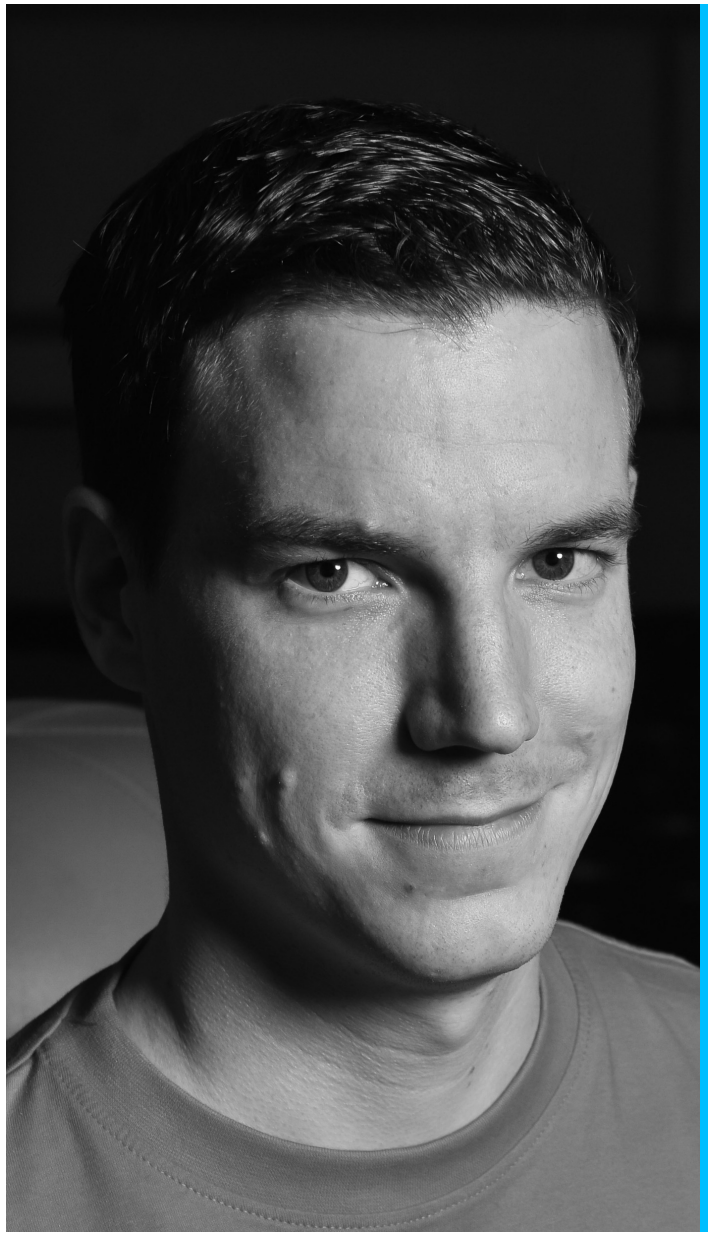
1. The requirement picks the framework — not the hype.

2. AI / Copilot is now one of the six lenses.

3. Modernize, don't rewrite. Then go build.

...and if you still ship WinForms. 😄

honestly? that's fine.



Thank you!

merci vielmol · grazie · danke

Now go pick one and ship it.

Slides + all 5 projects

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Microsoft MVP · iDIP Solution

iDIP Solution

Blazor · .NET MAUI · WinUI 3 · WPF · WinForms